

ACTIVITY 6

LEVELS:

Grade 12, University Preparation

COURSES:

Canada and World Issues

HE1.03

TASK:

Through a simulation exercise, students learn about the earth's fertility.

EXPECTATIONS:

The student will:

- Identify the finite nature of most of the world's resources.

TIMEFRAME:

15 Minutes

ACHIEVEMENT CATEGORIES:

Knowledge / Understanding

Thinking / Inquiry

Application / Making Connections

MATERIALS:

1 paring knife

1 apple

VOCABULARY:

Ecologically productive

Per capita quota

Genetically modified food

Crop rotation

Dykes



Apple of Our Eye

PROCEDURE:

This activity breaks the web of life developed in the introductory activity into its components and to see what the earth consists of.

1. Explain to the class that the apple represents the earth.
2. Slice your apple into four even parts.
3. Set three slices aside.
4. What do they represent? (The oceans)
5. Cut the remaining $\frac{1}{4}$ in half.
6. Set $\frac{1}{8}$ aside.
7. What does it represent? (Areas too inhospitable to most, except a few isolated groups: polar, desert, swamps, very high and rocky mountains).
8. What does the $\frac{1}{8}$ remaining represent? (Where people live but do not necessarily grow food).
9. Split this $\frac{1}{8}$ piece into four slices.
10. Set three aside.
11. What do they represent? ($\frac{3}{32}$ represent the areas too rocky, too cold, too steep or with soil too poor to produce food; also shopping centres, car parks, factories; roads and cities).
12. With the $\frac{1}{32}$ left, peel the skin.
(The skin represents the skin of the earth's crust which human kind depends on for food producing. This land is less than 2 metres deep.)

DISCUSSION ON THE APPLE OF OUR EYE

KNOWLEDGE / UNDERSTANDING

- How much land do you think the skin on the $\frac{1}{32}$ section of the apple represents?
(This $\frac{1}{32}$ piece of apple represents 16.06 billion acres of land that is ecologically productive.)
- What is the total world population and what does this reveal about the per capita quota of ecologically productive land?
(There are 6 billion people on the earth. This means that there are 2.67 acres of ecologically productive land per person.)
- According to your ecological footprint, how many acres of productive land are you consuming?

APPLICATION / MAKING CONNECTIONS

- What does this activity reveal to you about the world we live in?
(Appreciate the importance of maintaining intact the limited and finite amount of land available for food production. Advanced agricultural technology (GMF, biotechnology) increase yield but with a fixed land resource base and an increasing human population, each person's portion gets smaller.)

THINKING / INQUIRY

- What land use practices would you recommend for preserving the land that is ecologically productive? (Crop rotation, limiting the use of ecologically productive land that is used for grazing animals, using organic practices instead of pesticide use which eventually reduces the fertility of the soil.)
- Are you aware of any methods of increasing the percentage of land in the world that is ecologically productive?
(In mountainous regions, people build terraces to create new ground for growing food. In Holland, dykes have been built to protect the land and also to reclaim land from the sea for growing crops.)

HOMEWORK SUGGESTION (See Suggested Homework 3):

What measures are being taken by governments to ensure that this $\frac{1}{32}$ portion of the earth or 16.06 billion acres is preserved for future generations?

Give examples of government initiatives.
(e.g. Earth Summit, Brundtland Report)

**UNIT 2: THE APPLE OF OUR EYE
SUGGESTED HOMEWORK**

DATE: _____

NAME: _____



What measures are being taken by governments to ensure that this 1/32 portion of the earth or 16.06 billion acres is preserved for future generations? Find an example of a sustainable development strategy pertaining to agricultural practices and fill in the information required below.

Sustainable Development Strategy

What (Title):

Where (What country (ies)?):

When (Date):

Detail (What does it do?):

WEBSITE?